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Merchandise display security systems and methods

Abstract

Security systems and methods are provided. In one example, a security system includes at least one lock configured to protect one or more items from theft from the fixture, wherein the lock comprises a drive shaft configured to be moved between a latched position and an unlatched position, the fixture configured to be accessed in the unlatched position. The lock is configured to be moved between a locked state and an unlocked state for allowing the drive shaft to be moved between the latched position and the unlatched position when in the unlocked state, wherein the lock includes a cam sleeve having an internal cam surface configured to transition the lock between the locked state and the unlocked state in response to movement of the cam sleeve.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. application Ser. No. 17/825,802 filed on May 26, 2022, which claims the benefit of priority to U.S. Provisional Application No. 63/194,301, filed on May 28, 2021, the entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

(1) Embodiments of the present invention relate generally to security systems, locks, devices,

computer program products, and methods for protecting items from theft and/or the exchange of various types of information in a wireless network.

BACKGROUND OF THE INVENTION

(2) It is common practice for retailers to display relatively small, relatively expensive items of merchandise on a security device, such as a display hook or a display fixture, within security packaging commonly referred to as a “safer”, or otherwise on a display surface. The security device or safer displays an item of merchandise so that a potential purchaser may examine the item when deciding whether to purchase the item. The small size and relative expense of the item, however, makes the item an attractive target for shoplifters. A shoplifter may attempt to detach the item from the security device, or alternatively, may attempt to remove the security device from the display area along with the merchandise. Items of merchandise may also be secured using a display stand to allow users to sample the item for potential purchase. In some instances, the security device is secured to a display support using a lock operated by a key, for example, a mechanical lock. In other instances, the security device is secured to the display support using a lock operated by an electronic key to arm and disarm the security device.

BRIEF SUMMARY

(3) Embodiments of the present application are directed towards security systems and methods for protecting items from theft. In one embodiment a security system for a fixture is provided and includes at least one lock configured to protect one or more items from theft from the fixture. The lock comprises a drive shaft configured to be moved between a latched position and an unlatched position, and the fixture is configured to be accessed in the unlatched position. The lock is configured to be moved between a locked state and an unlocked state for allowing the drive shaft to be moved between the latched position and the unlatched position when in the unlocked state. In addition, the lock includes a cam sleeve having an internal cam surface configured to transition the lock between the locked state and the unlocked state in response to movement of the cam sleeve. In some cases, a computing device is provided and is configured to transmit a wireless authorization signal to the lock to transition the lock between the locked state and the unlocked state

(4) In another embodiment, a method for securing items from theft from a fixture is provided. The method includes providing at least one lock configured to protect one or more items from theft from the fixture, wherein the lock comprises a drive shaft and a cam sleeve. The method further includes causing the cam sleeve to move to transition the lock from a locked state to an unlocked state. In addition, the method includes causing the draft shaft to be moved between a latched position and an unlatched position while the lock is in the unlocked state, the fixture configured to be accessed in the unlatched position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a merchandise security system according to one embodiment of the present invention.

(2) FIG. 2 illustrates a merchandise security system according to another embodiment of the present invention.

(3) FIG. 3 illustrates a key in communication with a remote device via a cloud according to one embodiment.

(4) FIG. 4 illustrates a plurality of keys with different authorization levels according to one embodiment.

(5) FIG. 5 is a plan view of an electronic key according to one embodiment.

(6) FIG. 6 is a perspective view of the electronic key shown in FIG. 5.

(7) FIG. 7 is a plan view of an electronic key according to another embodiment.

- (8) FIG. 8 is a perspective view of the electronic key shown in FIG. 7.
- (9) FIG. 9 is a plan view of an electronic key according to another embodiment.
- (10) FIG. 10 is a perspective view of the electronic key shown in FIG. 9.
- (11) FIG. 11 is a perspective view of a merchandise security device according to one embodiment.
- (12) FIG. 12 is a perspective view of an electronic key according to one embodiment.
- (13) FIG. 13 is a cross-sectional view of the electronic key shown in FIG. 12.
- (14) FIG. 14 is a perspective view of a merchandise security device in a locked and unlocked position according to one embodiment.
- (15) FIG. 15 is a perspective view of a merchandise security device in a locked and unlocked position according to another embodiment.
- (16) FIG. 16 is a plan view of a charging station according to one embodiment.
- (17) FIG. 17 is a perspective view of the charging station shown in FIG. 16.
- (18) FIG. 18 illustrates a merchandise security system according to one embodiment.
- (19) FIG. 19 illustrates an electronic key in communication with a computing device according to one embodiment.
- (20) FIG. 20 illustrates top and bottom perspective views of an electronic key according to another embodiment.
- (21) FIG. 21 illustrates plan and side views of the electronic key shown in FIG. 20.
- (22) FIG. 22 is a plan view of a programming or authorization station according to one embodiment.
- (23) FIG. 23 is a perspective view of the programming or authorization station shown in FIG. 22.
- (24) FIG. 24 is another perspective view of the programming or authorization station shown in FIG. 22.
- (25) FIG. 25 is a schematic illustration of a plurality of sensors and alarm nodes communicating in a wireless network according to one embodiment.
- (26) FIG. 26 is a schematic of infrastructure and security devices within a wireless network according to one embodiment of the present invention.
- (27) FIG. 27 is a perspective view of a system in a wireless network according to one embodiment.
- (28) FIG. 28 is a perspective view of a system in a wireless network according to one embodiment.
- (29) FIG. 29 is a perspective view of a system in a wireless network according to one embodiment.
- (30) FIG. 30 is a perspective view of a system in a wireless network according to one embodiment.
- (31) FIG. 31 is a perspective view of a system in a wireless network according to one embodiment.
- (32) FIG. 32 shows various security devices configured for use in a wireless network according to additional embodiments.
- (33) FIG. 33 shows a security device configured for use in a wireless network according to one embodiment.
- (34) FIG. 34 shows a security device configured for use in a wireless network according to one embodiment.
- (35) FIG. 35 shows a security device configured for use in a wireless network according to one embodiment.
- (36) FIG. 36 shows a security device configured for use in a wireless network according to one embodiment.
- (37) FIG. 37 is a perspective view of a system in a wireless network according to one embodiment.
- (38) FIG. 38 is a perspective view of a system in a wireless network according to one embodiment.
- (39) FIG. 39 is a perspective view of a system in a wireless network according to one embodiment.
- (40) FIG. 40 is a perspective view of a system in a wireless network according to one embodiment.
- (41) FIG. 41 is a perspective view of a system in a wireless network according to one embodiment.
- (42) FIG. 42 is a perspective view of a system in a wireless network according to one embodiment.
- (43) FIG. 43 is a perspective view of a merchandise display security system according to one embodiment.

- (44) FIG. 44 illustrates various components of a merchandise display security system according to one embodiment.
- (45) FIGS. 45A-C illustrate internal can cross-sectional views of a lock according to one embodiment.
- (46) FIG. 46A-B are perspective views of different locks according to additional embodiments.
- (47) FIG. 47 is a perspective view of a merchandise display security system according to another embodiment.
- (48) FIG. 48 is a perspective view of a lock mounted to a fixture according to one embodiment.
- (49) FIG. 49 is a perspective view of a fixture having locks mounted thereto according to one embodiment.
- (50) FIGS. 50A-B are perspective views of different locks according to additional embodiments.
- (51) FIG. 51 is a perspective view of a lock and an electronic key according to one embodiment.
- (52) FIGS. 52A-B are perspective views of a lock having a modular component according to one embodiment.
- (53) FIG. 53A-C illustrate the operation of various locks according to additional embodiments.
- (54) FIGS. 54A-C illustrate the operation of various locks according to additional embodiments.
- (55) FIG. 55 is a perspective view of a lock according to one embodiment of the present invention.
- (56) FIG. 56 is a side view of the lock shown in FIG. 55.
- (57) FIG. 57 is an internal elevation view of the lock shown in FIG. 55.
- (58) FIG. 58 is an internal perspective view of the lock shown in FIG. 55.
- (59) FIG. 59 is a perspective cross-sectional view of the lock shown in FIG. 55.
- (60) FIG. 60 is an elevation cross-sectional view of the lock shown in FIG. 55.
- (61) FIG. 61 is another perspective cross-sectional view of the lock shown in FIG. 55.
- (62) FIG. 62 is an internal elevation view of the lock shown in FIG. 55.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

(63) The following disclosure includes various embodiments of systems, devices, methods, and computer program products. It should be understood that any combination of embodiments disclosed herein have been envisioned. Thus, discussion of one particular embodiment is not intended to be made at the exclusion of any other embodiments.

(64) Referring now to the associated figures, one or more embodiments of a security system are shown. In the embodiments shown and described herein, the system includes an electronic key and a merchandise security device. Examples of merchandise security devices suitable for use with the electronic keys include, but are not limited to, a security display (e.g. alarming stand or device), security fixture (e.g. locking hook, shelf, cabinet, etc.), cabinet locks, door locks, cable wraps, cable locks, or security packaging (e.g. merchandise keeper) for an item of merchandise. However, an electronic key (also referred to herein as a programmable key or generally as a key) may be useable with any security device or locking device that utilizes power transferred from the key to operate a mechanical and/or electronic lock mechanism and/or utilizes data transferred from the key to authorize the operation of a lock mechanism and/or arming or disarming an alarm circuit. In other words, an electronic key is useable with any security device or locking device that requires power transferred from the key to the device and/or data transferred from the key to the device. Further examples of security devices and locking devices include, but are not limited to, a door lock, a drawer lock or a shelf lock, as well as any device that prevents an unauthorized person from accessing, removing or detaching an item from a secure location or position. Although the following discussion relates to a system for use in a retail store, it is understood that the system is also suitable for other industries, such as hospital, restaurants, etc. In some embodiments, the merchandise security systems, merchandise security devices, and electronic keys are similar to those disclosed in U.S. application Ser. No. 17/668,931, entitled Merchandise Display Security Systems and Methods, PCT Publication WO 2020/227513 (and related U.S. application Ser. No. 17/261,757), entitled Merchandise Display Security Systems and Methods, U.S. Publication No.

2012/0047972, entitled Electronic Key for Merchandise Security Device, U.S. Pat. No. 10,258,172, entitled Systems and Methods for Acquiring Data from Articles of Merchandise on Display, U.S. Pat. No. 10,210,681, entitled Merchandise Display Security Systems and Methods, U.S. Publ. No. 2018/0365948, entitled Tethered Security System with Wireless Communication, and U.S. Publication No. 2016/0335859, entitled Systems and Methods for Remotely Controlling Security Devices, the entire disclosures of which are incorporated herein by reference in their entirety.

(65) FIG. 1 illustrates one embodiment of a system **10**. In this embodiment, the system generally includes an electronic key **12**, one or more merchandise security devices **14**, a programming or authorization station **16**, and a charging station **18**. FIG. 2 shows an embodiment of a system **10** that is part of a network of merchandise security devices. According to some embodiments, the network enables communication between a plurality of electronic keys and merchandise security devices. The network may be cloud-based and include a cloud **22** for receiving data from, and/or providing data to, the electronic keys and/or merchandise security devices. The cloud **22** may facilitate communication with one or more computing devices **26** (e.g., a mobile device, tablet, or computer). For example, the cloud **22** may be used to transfer data to one or more remote locations or computing devices **26** where the data may be reviewed and analyzed. The computing devices **26** may be located at any desired location, such as in the same retail store as the security devices **14** and/or electronic keys **12**. In some cases, the computing device **26** may belong to a retail store associate (e.g., a mobile device) or be a backend computer used by a retailer or corporation. The network may be a wireless network including a plurality of nodes **20** that are configured to communicate with one another, one or more electronic keys **12**, and/or one or more merchandise security devices **14**. The network may be any suitable network for facilitating wireless communication such as, for example, a mesh, star, multiple star, repeaters, IoT, etc. networks. The nodes **20** and/or security devices **14** may be located within one or more zones. In some cases, the nodes and the security devices may be integrated with one another such that the security device operates as a node. A gateway **24** or hub or “host” may be employed to allow for communication between the one or more nodes **20** and the cloud **22**. In some embodiments, all communication within the network is wireless, such as via radio-frequency signals (e.g., Sub GHz ISM band or 2.4 GHZ), Bluetooth, LoRa, and Wi-Fi, although other types of wireless communication may be possible.

(66) In some embodiments, each merchandise security device **14** and/or electronic key **12** is configured to store various types of data. For example, each merchandise security device **14** and/or key **12** may store a serial number of one or more merchandise security devices **14**, a serial number of one or more items of merchandise, the data and time of activation of the key, a user of the key, a serial number of the key, a location of the security device, a location of the item of merchandise, a department number within a retail store, number of key activations, a type of activation (e.g., “naked” activation, activation transferring only data, activation transferring power, activation transferring data and power), and/or various events (e.g., a merchandise security device has been locked, unlocked, armed, or disarmed). For instance, FIG. 3 shows that the identity of a user of an electronic key **12** may be communicated to a remote location or device **26**. This information may be transmitted to the remote location or device **26** upon each activation of the key **12** or at any other desired period of time, such as upon communication with a programming or authorization station **16**. Thus, the data transfer from the electronic key **12** and/or security device **14** may occur in real time or automatically in some embodiments. In some cases, the electronic key **12**, security device **14**, and/or programming station **16** may be configured to store the data and transfer the data to a remote location or device **26**. Authorized personnel may use this data to take various actions using the computing device **26**, such as to audit and monitor associate activity, authorize or deauthorize particular keys **12**, determine the battery life of a key **12**, audit merchandise security devices **14** (e.g., ensure the security devices are locked or armed), arm or disarm the security device, lock or unlock the security device, lock or unlock a sensor **25** attached to an item of

merchandise to a base or stand **35** removably supporting the sensor, etc. (see, e.g., FIG. **30**). Moreover, such information may be requested and obtained on demand using the computing device **26**, such as from the electronic keys **12**, security devices **14**, and/or the programming station **16**. (67) In some cases, the data may include battery analytics of an electronic key **12**. For example, the battery analytics may include monitoring the battery voltage of an electronic key **12** when the key is placed on a charging station **18** and the time taken to reach full charge. These values may be used to determine depth of discharge. The battery analytics may be indicative of a battery that is nearing its end of life. A retailer or other authorized personnel may take various actions using this information, such as replacing the key or disabling the key to prevent battery swelling and housing failure.

(68) In one embodiment, the electronic key **12** is configured to obtain data from a merchandise security device **14** (e.g., a security fixture). For example, the merchandise security device **14** may store various data regarding past communication with a previous electronic key **12** (e.g., key identification, time of communication, etc.), and when a subsequent electronic key communicates with the same merchandise security device, the data is transferred to the electronic key. Thus, the merchandise security device **14** may include a memory for storing such data. In some cases, the merchandise security device **14** includes a power source for receiving and storing the data, while in other cases, the power provided by the electronic key **12** is used for allowing the merchandise security device to store the data. The electronic key **12** may then communicate the data for collection and review, such as at a remote location or device **26**. In some instances, communication between the electronic key **12** and the programming or authorization station **16** may allow data to be pulled from the electronic key and communicated, such as to a remote location or device **26**. In other cases, the electronic key **12** may be configured to obtain data from merchandise security devices **14** (e.g., a security display), such as an identification of the merchandise security device, the type of item of merchandise on display, an identification of the item of merchandise, and/or the system health of the security device and/or the item of merchandise. The electronic key **12** may store the data and provide the data to a remote location or device **26** directly or upon communication with the programming or authorization station **16**. As such, the electronic keys **12** may be a useful resource for obtaining various types of data from the merchandise security devices **14** without the need for wired connections or complex wireless networks or systems.

(69) In one embodiment, the security device **14** may communicate its identifier using various techniques. For example, in some cases the security device **14** may have a memory configured to store a serial number and is able to communicate that serial number to the electronic key **12** using bi-directional communication. In instances where the security device **14** may not have a memory, power source, and/or the ability for bi-directional communication (e.g., a cable wrap or locking hook), the security device may have an RFID tag, an NFC tag, or the like that stores an identifier for the security device (e.g., a serial number). Such security devices may be similar to that disclosed in U.S. Pat. No. 9,133,649, entitled Merchandise Security Devices for Use with an Electronic Key, the entire disclosure of which is incorporated herein by reference in their entirety. In some examples, the tag may be attachable (e.g., via adhesive) to existing security devices **14** such that it is readily adaptable to current devices, or the tag may be integrated within the security device. The electronic key **12** may be configured to deliver power to the tag to read the identifier of the tag, such as for a passive tag, although the tags may be passive or active. The electronic key **12** may store a number of authorized identifiers in memory (e.g., via a look-up table) and may then determine if the read identifier is in its memory. Alternately, the electronic key **12** may be configured to wirelessly connect to a network device **26** with a look-up table. Either the electronic key **12** itself or the network device **26** can then determine if the particular key or user of that key is authorized to unlock the security device **14** with the read identifier. The identifier may be unique to the security device **14** or may be a more generic identifier, such as for example, a “6-sided box” or a department such as “healthcare” or all of the above. Once authorization has been obtained, only

then will the electronic key be capable of delivering power to the security device **14** to successfully operate the lock and unlock it. If there is no authorization, the electronic key **12** does not continue this cycle, and the lock never unlocks. Thus, embodiments of the present invention may be configured to communicate with any type of security device **14** for performing various auditing, zone control, and planogram analysis based on identification of the security device.

(70) In one embodiment, the electronic key **12** and security device **14** may communicate with one another via NFC to transmit data when the key and security device are positioned near one another or in direct contact with one another. An NFC tag may include various components, such as an antenna or a coil and one or more chips that define an electrical circuit. The antenna may be used for effectuating communication with an electronic key **12**, which may be activated via a magnetic field. For example, a magnetic field may be generated by the electronic key **12** to communicate with an NFC tag.

(71) In some embodiments where the electronic key **12** is configured to transfer power inductively, as explained in further detail below, and is equipped to communicate using NFC or RFID, the inductive coil of the key may be configured to use the same coil for both data transfer and power transfer. In some cases, the electronic key **12** is configured to switch the coil between an energy transfer mode and an NFC or RFID receiver circuit. In other examples, a plurality of security devices **14** may be “nested” with one another such that authorization to one of the nested security devices results in all security devices being disarmed or unlocked. For instance, a plurality of locks could be paired to one another such that successful communication between any one of the locks and the electronic key **12** results in all of the locks being unlocked.

(72) In some embodiments, the merchandise security devices **14** include wireless functionality for communicating within the network. For example, the merchandise security devices may communicate wirelessly with each other, items of merchandise, electronic keys **12**, computing devices **26**, and/or nodes, including but not limited to communicating the various types of data discussed herein. Thus, in some cases, the computing devices **26** may communicate directly with the security devices **14** and/or electronic keys **12**.

(73) One embodiment of such a wireless system includes various types of wireless networks capable of being used in conjunction with embodiments disclosed herein. In some cases, the wireless system includes fully integrated hardware, software, and data analytics which effectively eliminates or makes negligible the added hardware costs of a data integrated solution—all other features remaining constant. In some embodiments, the wireless system is configured to adapt to a changing market where an increasing number of smartphones leverage Qi based inductive charging and exposed data ports no longer exist. For instance, in an embodiment where the security device **14** includes a sensor **25** and a base or stand **35** (see, e.g., FIG. **30**), the sensor may utilize Qi technology, such as a Qi coil that is configured to communicate with a corresponding coil in the item of merchandise. In addition, embodiments of the wireless system may be configured to provide a common wireless interface and IP gateway for future networked products leveraging the various wireless networks discussed herein. Various modes of operation can be implemented according to wireless system embodiments. In one example, a non-IP connected mode could be employed whereby a customer choosing not to subscribe to a SaaS service is able to leverage the wireless system's display merchandising and security features independent of a connection to an IP enabled network. Another mode may include an IP-connected mode, which may provide information, e.g., regarding security armed and power status and alarm alerts alarm activity on a local store basis. Additionally, this mode may provide access to other web applications such as product documentation, product videos, product selector guides and support contact information. An additional mode is also an IP-connected network that includes a SaaS subscription service that allows access to the full capabilities of the wireless system, such as the data communication among various devices described herein.

(74) In some embodiments, wireless communication may occur using a proprietary wireless

network, for example, each security device **14** may be configured to communicate with a central hub in a star network configuration. Each security device **14** may include a transceiver (e.g., a sub-GHz transceiver) configured to communicate data to and from a common central hub or “host” **24**, such as the various types of information and data discussed herein, as well as information about power status and security breaches to the host without the need for a separate data connection to a smart hub or controller. It is understood that any number of nodes **20** could be employed to facilitate communication between the security devices **14** and the host, such one or more local nodes. In one embodiment, each security device **14** is configured to communicate its power and security status, security breaches (alarm notifications), as well as various other identification data for the security device and/or the item of merchandise, to the host **24**. In some embodiments, an entire retail store may be serviced by a single host **24** without the need for repeaters and is not practically limited by the number of security devices in the network. In one embodiment, the host **24** may be configured to generate a security signal, such as an audible and/or a visible alarm signal. In some cases, the volume of the security signal is adjustable. When any security device **14** detects a security event, the security device is configured to send a signal to the host **24**. The retailer has the option of choosing the level of notification for the security event, for example, a loud audible alarm, a lower volume, audible notification, or no audible alarm notification. Among other features, the system may include the ability to program alarm notifications. For instance, a retailer may choose silent alerts, optical alerts, and adjustable volume and tone audible alerts or combinations of these alerts. Additionally, the host **24** could be configured to indicate a security breach by changing colors (e.g., from gold to red and or by flashing intermittently). The audible and visual alert signals can be used independently or together.

(75) As discussed herein, electronic keys **12** may be incorporated with the various system embodiments. Electronic keys **12** may be configured to disable any alarming security device **14** following a security event. However, the host **24** may be configured to continue to transmit a security signal, such as until the security device **14** is re-armed. Moreover, disabling a security signal on the host **24** may not affect the armed status of the remaining security devices **14** in the store, i.e., the security devices may operate one-to-one in every regard except for generation of security signals. Of course, a variety of types of electronic keys **12** as disclosed herein, including leveraging a secure application available on a smartphone, tablet or PC.

(76) In some embodiments, a pre-emptive disarm for purposes of remerchandising items of merchandise or nightly removal of the item from an associated security device **14** may be employed. For example, a computing device **26** of the retailer (e.g., a mobile device) **26** may be configured to automatically disarm one or more security devices **14** at a predetermined period of time. In some cases, a secure software application may permit a temporary suspension of alerts for a specific position of a security device **14** for a programmable period to permit re-merchandising. Once disarmed, the security device's transceiver will cease communicating until it is re-armed. For those customers operating in a “Non-IP Connected” mode can elect to silence the audible alarm of the security device **14** when remerchandising such that no audible alarm will sound, but the host may continue to generate a signal (e.g., light signal) until all security devices are re-armed.

(77) As described herein, embodiments of the present invention may utilize a variety of wireless network configurations. In some cases, a common architecture would require two distinct network topologies. The first network may be a private wireless network for the exclusive use of the security devices **14** deployed instore. This network is separate from any private or public network operated by the retailer. The second network may be an IP Gateway between the private network and the Internet. This second network may be a connection on retailer's managed network or could be via a cellular modem. The gateway could be integrated into the host or be a separate device that connects to the host.

(78) In some embodiments, the private network may be commonly used by all security devices **14** for internal data transfer and minimize frequency congestion for retailer managed networks.

Moreover, in one example, the private network practically takes the form as a “star network”—with multiple individual nodes **20** performing individual functions and collecting and providing data. This data is wirelessly sent to and aggregated within a common “host”. The host allows nodes **20** providing data wirelessly via the private network to deliver functionality and value to the customer independent of an Internet connection to a cloud-based application, such as alerting and reporting functionality. In one implementation, the host rather than the security device **14** would be configured to provide notification (e.g., in response to a security event) via audio, visual, and/or haptic response.

(79) Various considerations may be taken into account regarding the private network. For instance, in selecting the appropriate, common network architecture for the private network, considerations of the size of the data packets and data rate required, the needed wireless range, potential for interference, power consumption, size, and/or cost of the network may be taken into account. In some applications, intermittent transmission of small data packets, with no need for higher data rates, may be used, which may benefit from a network with low power needs and long data range. Examples of private networks include various RF networks, such as Wi-Fi (2.4 GHZ), Bluetooth (2.4 GHZ) and Sub GHz (less than 1.0 GHZ) ISM band networks. Some network stacks (controlling software) such as Zigbee and LoRa can run on both sub GHz and 2.4 GHz networks.

(80) Another example embodiment of a wireless network system includes various types of security devices **14** and electronic keys **12** that may cooperate with one or more nodes **20**, hubs **24**, and/or computing devices **26** in a wireless network (see, e.g., FIGS. **26-42**). Various types of security devices **14** may be employed in the system, such as those disclosed herein. For example, security devices **14** that include a sensor that is configured to be attached to an item (e.g., via adhesive and/or brackets). In some implementations, the sensor may be connected to a base or stand **35** with a tether **45** (see, e.g., FIGS. **30-32**), or no tether may be used in some cases (see, e.g., FIGS. **32-33**). Sensors **25** may take many different forms, such as, for example, standalone sensors (see, e.g., FIG. **36**), “chairback” sensors (see, e.g., FIG. **33**), sensors that provide power and security for the item of merchandise (e.g., via USB-C, micro-USB, etc. connectors) (see, e.g., FIG. **35**), and/or sensors that only provide security (e.g., a sensor including a plunger switch) (see, e.g., FIG. **34**). Similarly, the base **35** used to removably support a sensor **25** may also take different forms (see, e.g., FIG. **33** where a chairback sensor is used with electrical contacts for transferring power between the sensor and the base). Of course, the security devices **14** may be used in various industries such as retail stores and for a variety of items, such as merchandise or commercial items (e.g., tablet computers).

(81) As shown in FIGS. **27-29**, various numbers and types of security devices **14** may be configured to communicate with one another in a network, such as a private wireless network as discussed above. A host or hub **24** may be configured to communicate with each of the plurality of security devices **14** in the network and provide various security signals, such as disclosed herein. An interface may be provided on the hub **24** for facilitating communication with an electronic key **12**. FIG. **27** shows an example where the plurality of security devices **14** and hub **24** are configured to communicate in an IP network which may allow for various information and alerts to be provided to one or more computing devices **26** (e.g., system health, power status, alarm status, and/or inventory information). Moreover, FIG. **28** illustrates an example similar to FIG. **27** but where the system includes additional features via a SaaS subscription to enterprise software, such as for example, displaying planogram (“POG”) compliance information, consumer activity, programmable KPI's, inventory re-stock thresholds, and/or inventory POG compliance. FIGS. **30-31** show various depictions of a plurality of security devices **14** in the form of a sensor and base which are configured to communicate with a hub **24** and a computing device **26** configured to receive notifications from the hub (e.g., no power at the security device or a breach has occurred). Furthermore, FIGS. **37-42** illustrate embodiments of security devices **14** in the form of locks that are configured to communicate in the wireless network with the hub **24**. In these examples, a customer may be able to request assistance (e.g., via a call button on the security device **14**) that

enables a sales associate to be notified and to thereafter engage the customer or control the security device **14** with an electronic key **12** or computing device **26**. The retail associate could use an electronic key **12** to unlock the security device **14** for the customer (see, e.g., FIG. **38**), or use a computing device **26** to unlock the security device. In some cases, the customer's mobile telephone may perform some of the functions disclosed herein ("Trusted Customer"), such as unlocking a security device **14** in response to receiving a wireless authorization signal (see, e.g., FIG. **39**). For example, a Trusted Customer may be a customer who has purchased an item and is picking the item up in the store or one who has an account with the retailer and is purchasing the item using the customer's mobile device. In addition, various data may be collected regarding the security device **14**, such as for example, the type of product that was removed from a cabinet or drawer protected by a lock, and allows for alerts to be provided to one or more computing devices **26** (see, e.g., FIG. **40**). The security devices **14** may be configured to automatically relock after an authorized opening and accessing the item of merchandise (see, e.g., FIG. **41**), and various techniques may be employed to track items of merchandise added or removed from a cabinet or drawer, such as an RFID scanner that is configured to scan the product as the item is added or removed from the cabinet or drawer (see, e.g., FIG. **42**).

(82) In other embodiments, inventory information may be obtained regarding merchandise on a security device **14** such as a locking hook, information may be obtained regarding items of merchandise removed from a security device (e.g., a cabinet), and computing devices **26** may be used to obtain various types of information and provide various types of commands for controlling the security device and/or item of merchandise. Embodiments of wireless systems disclosed herein may provide for real time reporting of Who/What/When/Where/Why/How for interactions with security devices **14** and items of merchandise, be responsive/interactive, migrate from security focus to omni-channel experience enablement within the retail store, facilitate Trusted Customer engagement with security assets, allow to readily customize and expand the system, enable alternative business models such as SaaS models, connect local network of connected assets with central hub for local computing, and/or connect hub to cloud platform for providing alerts, reporting, system administration, daily operation. Embodiments may also provide a platform infrastructure having a centralized hub per retail store and several fit for purpose connected end security device assets such as stands, sensors, table managers, locks, cabinet sensors, inventory sensors, customer dwell sensors, etc. that all communicate with the hub. Due to the flexibility of wireless systems in some embodiments, customers do not need to pre-select which security devices **14** to purchase since the platform infrastructure is common. Furthermore, computing devices **26** and mobile devices used by retailers may allow retailers and store associates to dynamically interact with security devices **14** to make real-time decisions, such as responding to security events, restocking out of stock inventory, or responding to customer requests for assistance with secured items of merchandise.

(83) In some cases, each electronic key **12** may be authorized for specific locations, departments, or merchandise security devices. For instance, FIG. **4** shows that a manager may have authorization for all zones, locations, departments, or merchandise security devices (indicated as numbers 1-6), while a first associate may only have authorization for two zones, locations, departments, or merchandise security devices (indicated as numbers 4 and 5), and a second associate may only have authorization for one zone, location, department, or merchandise security device (indicated as number 6). As such, a retail store or other establishment may limit the scope of authorization for different associates within the same retail store. In order to accommodate different authorizations levels, each key **12** may be configured to store a code that is associated with each zone, location, department, or merchandise security device. For example, each zone may include a plurality of merchandise security devices **14**, and a retail store may have multiple zones (e.g., a zone for electronics, a zone for jewelry, etc.).

(84) Various techniques may be used to initially program the electronic key **12**. For example, the

electronic key **12** may be initially presented to each authorized merchandise security device **14**. Upon communication with the security device **14** or the cloud **22**, the electronic key **12** will be paired with each security device. A programming station **16** may provide a code to the electronic key **12**, and the key or cloud **22** may then communicate the code to each of its authorized security devices **14**. Each key **12** may only need to be programmed once. In some embodiments, a programming station **16** may be located within each zone, and a key **12** may receive a code from each programming station that it is authorized. Thereafter, each key **12** may need to be “refreshed” at the programming station **16** or a charging station **18** following a predetermined period of time or in response to being disabled as described in various examples herein. In other embodiments, the electronic key **12** may be programmed directly via the cloud **22**.

(85) In another embodiment, each electronic key **12** may include a security code and a serial number for one or more merchandise security devices **14**. For example, a key **12** may only be able to arm, disarm, lock, or unlock a merchandise security device **14** where the security codes and the serial numbers match one another. In one example, each serial number is unique to a merchandise security device **14** and could be programmed at the time of manufacture or by the retailer. This technique allows for greater flexibility in programming keys **12** and assigning keys to particular merchandise security devices **14** and/or zones. In one embodiment, a setup electronic key **12''** may be used to initially map particular merchandise security devices **14** and serial numbers. In this regard, the setup key **12''** may be used to communicate with each key **12** and obtain the serial number of each merchandise security device **14**. The setup key **12''** may also obtain a location of the security devices **14**, or a user of the setup key may provide a description for each merchandise security device (e.g., SN #123=merchandise security device #1). The setup key **12''** may communicate with a tablet or other computing device **26** for accumulating all of the information (see, e.g., FIGS. **3** and **19**), which may occur via wired or wireless communication. Thus, the tablet or computing device **26** may map each of the serial numbers with the merchandise security devices **14** and in some cases, may also include serial numbers and corresponding electronic keys **12**. Individual electronic keys **12** may then be assigned particular serial numbers for authorized merchandise security devices **14** (e.g., user **1** includes serial numbers 1, 2, 3; user **2** includes serial numbers 1, 4, 5). Each of the electronic keys **12** may be programmed with the same security code using a programming station **16**. In some embodiments, the setup process may be used in conjunction with a planogram of the merchandise security devices **14**. The planogram may represent a layout of the merchandise security devices **14** within a retail store or other establishment. For example, a setup key **12''** may be used to map serial numbers to specific merchandise security devices **14** on a planogram as the setup key communicates with each merchandise security device. The setup key **12''** may communicate with a tablet or other computing device **26** for populating the planogram with serial numbers, such as via a wired connection (see, e.g., FIG. **19**). This planogram may be uploaded to a remote location or device for managing the planogram and ensuring planogram compliance based on information exchanged between the security devices **14** and the computing device **26**. As before, particular serial numbers may be assigned to authorized users.

(86) In order to arm, disarm, lock, or unlock a merchandise security device **14**, the electronic key **12** may communicate with a particular merchandise security device and determine whether the security codes and the serial numbers match. If the codes match, the electronic key **12** then arms, disarms, locks, or unlocks the merchandise security device **14**. Upon refreshing an electronic key **12** and/or when a user requests an electronic key via programming or authorization station **16**, any available electronic key may be used since the key may be programmed in real time with the appropriate level of authorization for that user (e.g., specific zones, departments, and/or merchandise security devices).

(87) In one embodiment, the merchandise display security system **10** comprises an electronic key **12** and a merchandise security device **14** that is configured to be operated by the key. The system

may further comprise an optional programming station **16** that is operable for programming the key **12** with a security code, which may also be referred to herein as a Security Disarm Code (SDC). In addition to programming station **16**, the system may further comprise an optional charging station **18** that is operable for initially charging and/or subsequently recharging a power source disposed within the key **12**. For example, the key **12** and merchandise security device **14** may each be programmed with the same SDC into a respective permanent memory. The key **12** may be provisioned with a single-use (i.e., non-rechargeable) power source, such as a conventional or extended-life battery, or alternatively, the key may be provisioned with a multiple-use (i.e. rechargeable) power source, such as a conventional capacitor or rechargeable battery. In either instance, the power source may be permanent, semi-permanent (i.e., replaceable), or rechargeable, as desired. In the latter instance, charging station **18** is provided to initially charge and/or to subsequently recharge the power source provided within the key **12**. Furthermore, key **12** and/or merchandise security device **14** may be provided with only a transient memory, such that the SDC must be programmed (or reprogrammed) at predetermined time intervals. In this instance, programming station **16** is provided to initially program and/or to subsequently reprogram the SDC into the key **12**. As will be described, key **12** may be operable to initially program and/or to subsequently reprogram the merchandise security device **14** with the SDC. Key **12** is then further operable to operate the merchandise security device **14** by transferring power and/or data to the device, as will be described.

(88) In the exemplary embodiment of the system illustrated in FIGS. **1-2**, electronic key **12** is configured to be programmed with a unique SDC by the programming station **16**. In some embodiments, the key **12** is presented to the programming station **16** and communication therebetween is initiated, for example, by pressing or otherwise actuating a control button **28** provided on the exterior of the key. Communication between the programming station **16** and the key **12** may be accomplished directly, for example by one or more electrical contacts, or indirectly, for example by wireless communication. Any form of wireless communication capable of transferring data between the programming station **16** and key **12** is also possible, including without limitation optical transmission, acoustic transmission or magnetic induction. In some embodiments shown and described herein, communication between programming station **16** and key **12** is accomplished by wireless optical transmission, and more particularly, by cooperating infrared (IR) transceivers provided in the programming station and the key. In some embodiments, the programming station **16** may function similarly to that disclosed in U.S. Pat. No. 7,737,844 entitled PROGRAMMING STATION FOR A SECURITY SYSTEM FOR PROTECTING MERCHANDISE, the disclosure of which is incorporated herein by reference in its entirety. For the purpose of describing some embodiments of the present invention, it is sufficient that the programming station comprises at least a logic control circuit for generating or being provided with a SDC, a memory for storing the SDC, and a communications system suitable for interacting with the electronic key **12** in the manner described herein to program the key with the SDC.

(89) An available feature of a merchandise security system **10** according to one embodiment is that the electronic key **12** may include a time-out function. More particularly, the ability of the key **12** to transfer data and/or power to the merchandise security device **14** may be deactivated after a predetermined time period. By way of example, the electronic key **12** may be deactivated after about six to about twenty-four hours from the time the key was programmed or last refreshed. In this manner, an authorized sales associate typically must program or refresh the key **12** assigned to him at the beginning of each work shift. Furthermore, the charging station **18** may be configured to deactivate the electronic key **12** when the key is positioned within or otherwise engaged with a charging port **30** (see, e.g., FIG. **1**). In this manner, the charging station **18** can be made available to an authorized sales associate. In one embodiment, the electronic key **12** may be authorized upon the sales associate inputting an authorized code to release the key for use. For instance, the sales associate may input a code on a keypad in communication with the charging station **18**. Upon

inputting the correct code, the charging station **18** may indicate which key **12** is authorized for use by the sales associate (e.g., via an audible and/or a visible indicator). In some cases, the time-out period may be predetermined or customized by a user. For example, a manager of a retail store may input a particular time period for one or more of the electronic keys **12**. Those electronic keys **12** that are “active” may be monitored via communication within the cloud-based network. In other embodiments, the electronic key **12** may be timed out or otherwise disabled in response to an event. For instance, the electronic key **12** may be disabled in response to the key being misplaced or stolen, or keys being brought into a retail store that are not authorized for use. Such disabling may alternatively occur via a command from a device **26** sent to the electronic key **12** via the cloud **22**. In other cases, the electronic key **12** may be disabled in response to failure to communicate with the network (e.g., at a particular time or time interval), a lost connection to the network, and/or an inability to reconnect to the network. In another example, the electronic key **12** may be disabled in response to its memory being full, e.g., with audit data.

(90) In one embodiment, commands may be provided remotely for taking various actions. For example, where a theft has occurred, a command may be provided from a remote location or device **26** (e.g., a tablet or computer) to lock and/or arm all or a portion of the merchandise security devices **14**. Similarly, a command may be provided from a remote location or device **26** to deactivate all or a portion of the electronic keys **12** and/or security devices **14**. As such, the system **10** provides techniques for centralized security and control of the electronic keys **12**, merchandise security devices **14**, and other components within the system. As discussed above, the electronic keys **12** may also be controlled remotely. Furthermore, in some embodiments, such requests or commands may be made by the computing device **26** for individual security devices **14** or a plurality of security devices (e.g., sending a command to lock all security devices in response to a security event). Moreover, one or more of the security devices **14** may be configured to lock or alarm in response to a security event (e.g., automatically locking a sensor attached to an item of merchandise to a base removably supporting the sensor).

(91) FIGS. 5-6 illustrate one embodiment of an electronic key **12**. The electronic key **12** may include a control button **28** for activating the key, such as for initiating communication with a merchandise security device. Moreover, the electronic key **12** may also include one or more visual indicators. In this regard, the key **12** may include one or more status indicators **32** that illustrate a status of the communication of the key with a merchandise security device **14**. The status indicators **32** may guide the user to know when communication between the key **12** and the merchandise security device **14** is taking place and has been completed. The status indicators **32** may be different depending on whether the communication was authorized (e.g., unlocked or disarmed), unauthorized (e.g., wrong zone or department), or unsuccessful. The status indicators **32** may also indicate an amount of time of authorized use remaining on the key **12**, such as where the key includes a time-out feature as discussed above. The electronic key **12** may also include one or more other indicators **34** that provide a visual indication of the power remaining on the key. These other indicators **34** may also be used for any other desired purpose, such as to indicate a programming state of the key **12**. For example, the indicators **34** may be activated while the electronic key **12** is being initially programmed. It is understood that the illustrated status indicators **32**, **34** are for illustration only, as various types and configurations of indicators may be employed in alternative embodiments.

(92) FIGS. 7-10 illustrate additional embodiments of electronic keys **12**. In these examples, the electronic key **12** includes a removable portion **36**. In FIGS. 7-8, the removable portion **36** allows access to an input power port **38**, such as for recharging the electronic key **12**. The removable portion **36** may be configured to slide relative to the electronic key **12** to expose the input power port **38**. The input port **38** may be configured to receive and electrically connect to a corresponding connector, such as a connector associated with the charging station **18**. For instance, the electronic key **12** may be configured to be docked within the charging station **18** for charging thereof (see,

e.g., FIG. 1). As shown in FIGS. 9-10, the removable portion 36 may also be configured to be removed entirely from the electronic key 12 and may be multi-purpose in that it may include a tool portion 40. For example, the tool portion 40 may be used for facilitating the disconnection of various connectors, as a screwdriver, etc. The electronic key 12 may include an opening 42 defined to receive the removable portion 36 therein in a non-use position.

(93) FIGS. 20-21 show additional embodiments of an electronic key 12'. In this embodiment, the electronic key 12' includes one or more alignment features 15 for facilitating alignment with a programming or authorization station 16' and/or a charging station 18' as discussed in further detail below. In addition, the electronic key 12' includes an input port 17 (e.g., a micro-USB port) which may be configured to releasably engage a corresponding port on the programming or authorization station 16' and/or the charging station 18' for data and/or power transfer. Notably in the example shown in FIG. 20, the input port 17 on the electronic key 12' is on a side surface, while a pair of alignment features 15 are provided on opposite surfaces of the electronic key. In the embodiment shown in FIG. 21, a single alignment feature 15 is provided. The input port 17 may be located on a side surface between a transfer port at one end and a key chain ring opening at an opposite end. Positioning of the input port 17 on a side surface of the electronic key 12' may provide for a more secure and stable attachment to the programming or authorization station 16' and/or the charging station 18'. A series of status indicators 32, 34, as discussed above, for example light-emitting diodes (LEDs) may be provided on the exterior of the electronic key 12' for indicating the operating status thereof.

(94) As shown in FIG. 1, the programming station 16 comprises a housing configured to contain the logic control circuit that generates the SDC, the memory that stores the SDC, and a communications system for communicating the SDC to the key (e.g., wirelessly). In use, the logic control circuit generates the SDC, which may be a predetermined (i.e. "factory preset") security code, a manually input security code, or a security code that is randomly generated by the logic control circuit. In the latter instance, the logic control circuit further comprises a random number generator for producing the unique SDC. A series of visual indicators, for example light-emitting diodes (LEDs) may be provided on the exterior of the housing for indicating the operating status of the programming station 16. Programming station 16 may further be provided with an access mechanism for preventing use of the programming station by an unauthorized person. For example, the programming station may include a keypad 44. An authorized user may input a code in the keypad 44 that allows the programming station 16 to generate a SDC for communicating to the key 12.

(95) In a particular embodiment, the logic control circuit of the programming station 16 performs an electronic exchange of data with a logic control circuit of the key, commonly referred to as a "handshake communication protocol." The handshake communication protocol determines whether the key 12 is an authorized key that has not been programmed previously (e.g., a "new" key), or is an authorized key that is being presented to the programming station 16 a subsequent time to refresh the SDC. In the event that the handshake communication protocol fails, the programming station 16 will not provide the SDC to the unauthorized device attempting to obtain the SDC. When the handshake communication protocol succeeds, programming station 16 permits the SDC to be transmitted by the key 12. As will be readily apparent to those skilled in the art, the SDC may be transmitted from the programming station 16 to the key 12 by any suitable means, including without limitation, wireless, electrical contacts or electromechanical, electromagnetic or magnetic conductors, as desired. Moreover, in other cases the programming station 16 may simply provide the SDC to the electronic key 12 without first initiating any handshake communication protocol.

(96) In some embodiments, the merchandise security device 14 is a "passive" device. As used herein, the term passive is intended to mean that the security device 14 does not have an internal power source sufficient to lock and/or unlock a mechanical lock mechanism. Significant cost savings are obtained by a retailer when the merchandise security device 14 is passive since the expense of an internal power source is confined to the key 12, and one such key is able to operate

multiple security devices. If desired, the merchandise security device **14** may also be provided with a temporary power source (e.g., capacitor or limited-life battery) having sufficient power to activate an alarm, for example a piezoelectric audible alarm, that is actuated by a sensor, for example a contact, proximity or limit switch, in response to a security breach. The temporary power source may also be sufficient to communicate data, for example a SDC, from the merchandise security device **14** to the key **12** to authenticate the security device and thereby authorize the key to provide power to the security device. In other cases, the security device may be an electronic device, such as a sensor attached to the item of merchandise and a base that removably supports the sensor thereon. The sensor may be attached to the base with a tether or may be wireless (e.g., using ranging techniques as described in more detail below).

(97) In some embodiments, the merchandise security device **14** further comprises a logic control circuit, similar to the logic control circuit disposed within the key **12**, adapted to perform a handshake communication protocol with the logic control circuit of the key in essentially the same manner as that between the programming station **16** and the key. In essence, the logic control circuit of the key **12** and the logic control circuit of the merchandise security device **14** communicate with each other to determine whether the merchandise security device is an authorized device that does not have a security code, or is a device having a matching SDC. In the event the handshake communication protocol fails (e.g., the device is not authorized or the device has a non-matching SDC), the key **12** will not program the device with the SDC, and consequently, the merchandise security device will not operate. If the merchandise security device **14** was previously programmed with a different SDC, the device will no longer communicate with the key **12**. In the event the handshake communication protocol is successful, the key **12** permits the SDC stored in the key to be transmitted to the merchandise security device **14** to program the device with the SDC. As will be readily apparent to those skilled in the art, the SDC may be transmitted from the key **12** to the merchandise security device **14** by any suitable means, including without limitation, via radiofrequency, one or more electrical contacts, electromechanical, electromagnetic or magnetic conductors, as desired. Furthermore, the SDC may be transmitted by inductive transfer of data from the electronic key **12** to the merchandise security device **14**. Moreover, in other cases the electronic key **12** may simply provide the SDC to the merchandise security device **14** without first initiating any handshake communication protocol.

(98) In one embodiment, when the handshake communication protocol is successful and the merchandise security device **14** is an authorized device having the matching SDC, the merchandise security device may be armed or disarmed, such as where the security device includes an alarm circuit. In other embodiments, the merchandise security device **14** may be armed or disarmed when the SDC codes match. In some embodiments, when the handshake communication protocol is successful and the SDC codes match, the logic control circuit of the key **12** causes an internal power source of the key to transfer electrical power to the device **14** to operate a mechanical lock mechanism. In other embodiments, the merchandise security device **14** may be locked or unlocked when the SDC codes match and power is transferred to the merchandise security device. It is understood that various information and codes may be exchanged in order to perform the desired function, such as arming, disarming, locking, or unlocking the merchandise security device **14**. For example, the data exchanged may include a serial number of the merchandise security device alone and/or an SDC.

(99) FIG. **11** shows one embodiment of a merchandise security device **140** in greater detail. As previously mentioned, the merchandise security device **14** can be any type of security device that utilizes an alarm circuit and/or a lock mechanism that locks and/or unlocks a lock. In some cases, the merchandise security device **140** may be a passive device in the sense that it does not have an internal power source sufficient to operate a lock mechanism. As a result, the merchandise security device **140** may be configured to receive power, or alternatively, both power and data, from an external source, such as the electronic key **12** shown and described herein. The embodiment of the

merchandise security device depicted in FIG. 11 is a cabinet lock configured to be securely affixed to the locking arm **104** of a conventional cabinet lock bracket **105**. As previously described, the cabinet lock **140** may include a logic control circuit for performing a handshake communication protocol with the logic control circuit of the key **12** and for receiving the SDC from the key. In other embodiments, the cabinet lock **140** may be configured to transmit the SDC to the key **12** to authenticate the security device and thereby authorize the key to transfer power to the security device.

(100) FIG. 12 shows an embodiment of an electronic key **120** with inductive transfer in greater detail. As previously mentioned, the key **120** may be configured to transfer both data and power to a merchandise security device **140**. Accordingly, the programmable electronic key **120** may be an active device in the sense that it has an internal power source sufficient to operate a mechanical lock mechanism of the merchandise security device **140**. As a result, the programmable electronic key **120** may be configured to transfer both data and power from an internal source, such as a logic control circuit (e.g., data) and a battery (e.g., power) disposed within the key. The embodiment of the programmable electronic key **120** depicted herein is a key with inductive transfer capability configured to be received within a transfer port **142** of the cabinet lock **140** shown in FIG. 11, as well as a programming port **46** of the programming station and the charging port **30** of the charging station. Thus, the electronic key **120** may be placed proximate to or within the transfer port **142** for communicating therewith. In some embodiments, a tag (e.g., RFID or NFC tag) as discussed above, may be positioned within the transfer port, or otherwise on the security device **140**, so that the electronic key **120** is configured to read or otherwise obtain identification data from the tag.

(101) In some embodiments, the electronic key **120** comprises a housing **121** having an internal cavity or compartment that contains the internal components of the key, including without limitation the logic control circuit, memory, communication system and battery, as will be described. As shown, the housing **121** is formed by a lower portion **123** and an upper portion **124** that are joined together after assembly, for example by ultrasonic welding. The electronic key **120** further defines an opening **128** at one end for coupling the key to a key chain ring, lanyard or the like. The electronic key **120** may further comprise a transfer probe **125** located at an end of the housing **121** opposite the opening **128** for transferring data and/or power to the merchandise security device **140**. The transfer probe **125** is also operable to transmit and receive a handshake communication protocol and the SDC from the programming station **16**, as previously described, and to receive power from a charging station.

(102) As best shown in FIG. 13, an internal battery **131** and a logic control circuit, or printed circuit board (PCB) **132** are disposed within the housing **121** of the electronic key **120**. Battery **131** may be a conventional extended-life replaceable battery or a rechargeable battery suitable for use with the charging station **18**. The logic control circuit **132** is operatively coupled and electrically connected to a switch **133** that is actuated by the control button **122** provided on the exterior of the key **120** through the housing **121**. Control button **122** in conjunction with switch **133** controls certain operations of the logic control circuit **132**, and in particular, transmission of the data and/or power. In that regard, the logic control circuit **132** is further operatively coupled and electrically connected to a communication system **134** for transferring data and/or power. In one embodiment, the communication system **134** is a wireless infrared (IR) transceiver for optical transmission of data between the electronic key **120** and the programming station, and between the key and the merchandise security device **140**. As a result, the transfer probe **125** of the key **120** may be provided with an optically transparent or translucent filter window **135** for emitting and collecting optical transmissions between the key **120** and the programming station **16**, or between the key and the merchandise security device **140**, as required. Transfer probe **125** may further comprise an inductive core **127** and inductive core windings **129** for transferring electrical power to the merchandise security device **140** and/or receiving electrical power from the charging station **18** to charge the internal battery **131**, as required. Alternatively, the optical transceiver **134** may be

eliminated and data transferred between the programmable electronic key **120** and the merchandise security device **140** via magnetic induction through the inductive coil **126**.

(103) In some embodiments, an important aspect of an electronic key **120**, especially when used for use in conjunction with a merchandise security device **140** as described herein, is that the key does not require a physical force to be exerted by a user on the key to operate the mechanical lock mechanism of the merchandise security device. By extension, no physical force is exerted by the key **120** on the mechanical lock mechanism. As a result, the key **120** cannot be unintentionally broken off in the lock, as often occurs with conventional mechanical key and lock mechanisms. Furthermore, neither the key **120** nor and the mechanical lock mechanism suffer from excessive wear as likewise often occurs with conventional mechanical key and lock mechanisms. In addition, in some cases there is no required orientation of the transfer probe **125** of the electronic key **120** relative to the ports on any one of the programming station, charging station, and/or the merchandise security device **140**. Accordingly, any wear of the electrical contacts on the transfer probe **125** and ports may be minimized. As a further advantage in some embodiments, an authorized person is not required to position the transfer probe **125** of the electronic key **120** in a particular orientation relative to the transfer port **142** of the merchandise security device **140** and thereafter exert a compressive and/or torsional force on the key to operate the mechanical lock mechanism of the device.

(104) FIGS. **22-24** illustrate an embodiment of a programming or authorization station **16'**. As illustrated, the programming or authorization station **16'** includes a geometry for receiving the electronic key **12'** as discussed above (see, e.g., FIG. **21**). In this regard, the programming or authorization station **16'** may include one or more alignment features **15'** configured to align with and engage alignment feature **15** of the electronic key **12'**. Moreover, the programming or authorization station **16'** may further define a recess **48** for at least partially receiving a side surface of the electronic key **12'**. The recess **48** may be curved or any other shape for corresponding to the shape of the electronic key **12'**. Within the recess **48**, the programming or authorization station **16'** may include a port **30'** for releasably engaging the input port **17** of the electronic key **12'**. The alignment features **15, 15'** are configured to align with one another to ensure that the input port **17** and port **30'** align with and engage one another. Such engagement may allow for data communication between the electronic key **12'** and the programming or authorization station **16'**, which may occur in some cases, upon entry of an authorized code using keypad **44**. In addition, the programming or authorization station **16'** may include one or more input ports **50** for receiving power and data communication (e.g., an Ethernet port).

(105) FIG. **1** shows a charging station **18** in greater detail. As previously mentioned, the charging station **18** recharges the internal battery **131** of the key **12**. In certain instances, the charging station **18** also deactivates the data transfer and/or power transfer capability of the key **12** until the key has been reprogrammed with the SDC by the programming station **16** or the user provides an authorized code to the charging station. Regardless, the charging station **18** comprises a housing for containing the internal components of the charging station. The exterior of the housing has at least one, and preferably, a plurality of charging ports **30** formed therein that are sized and shaped to receive the electronic key **12** (see, e.g., FIG. **1**). Mechanical or magnetic means may be provided for properly positioning and securely retaining the key **12** within the charging port **18** for ensuring proper power transfer.

(106) FIGS. **16-18** show an embodiment of a charging station **18** wherein a plurality of ports **30** are provided for engagement with a plurality of corresponding electronic keys **12'**. The electronic key **12'** shown in FIG. **21** may be compatible with the charging station **18** shown in FIGS. **16-18** whereby the electronic key **12'** includes an input port **17** on its side for engagement with the port **30**, similar to that described in conjunction with programming or authorization station **16'**. Likewise, each port **30** may be located within a respective recess **48** for receiving at least a side surface of the electronic key **12'**. This arrangement may allow for a greater number of electronic

keys **12'** to be engaged with the charging station **18** at any one time.

(107) FIGS. **14-15** show additional embodiments of a merchandise security device **150**. In this embodiment, the merchandise security device **150** comprises a lock mechanism that utilizes “energy harvesting”. Thus, the merchandise security device **150** may be a passive device as described above. However, in this embodiment, the merchandise security device **150** includes means for generating power to be stored. For example, the merchandise security device **150** may be configured to rotate between locked and unlocked positions and include a generator configured to generate energy to be stored (e.g., via a capacitor). In some cases, the merchandise security device **150** may include a bezel and each turn of the bezel may generate an electrical charge to be stored. In one embodiment, the electronic key **12** may be used initially to disengage a mechanical lock, and then the merchandise security device **150** may be rotated to an unlocked position. The merchandise security device **150** may then be rotated back to the locked position. Since the merchandise security device **150** has no power source, the security device is capable of performing various security functions using the stored power. For instance, the merchandise security device **150** may be configured to use the stored power to push data to one or more nodes **20** or to generate audible and/or visible signals. In one example, the merchandise security device **150** may include an internal radio for transmitting wireless signals using the stored power, such as for generating a distress signal when the security device is tampered with. In another example, the merchandise security device **150** may include a light-emitting device (LED) that is powered by the stored power.

(108) In another embodiment, a plurality of nodes are employed for peer-to-peer communication to facilitate the generation of an alarm signal, such as audible and/or visible signals. For example, FIG. **25** shows a plurality of merchandise security devices **14** (e.g., sensors) and alarm nodes **30** configured to wirelessly communicate various information to a gateway **24** via a network. For example, the sensors **14** and/or nodes **30** may be configured to send information to and receive information from the gateway **24** regarding their configuration, alarm status (e.g., alarming, armed, disarmed), and/or instructions (e.g., arm, alarm, or disarm). The merchandise security devices **14** and nodes **30** may also be configured to communicate directly with one another as described below, as well as to switch between communication with the gateway **24** and one another. Any number of nodes **30** could be located at various positions within a retail store, for example, such as on a display table or store entrance or exit. The nodes **30** may communicate wirelessly with merchandise security devices **14** and a gateway **24** within a network, such as described above using various wireless communication protocols. One disadvantage of using wireless communication to initiate the alarm at a location that is remote from the merchandise security device **14** is that the alarm signals often have to travel to a wireless hub where a server then deciphers the data and decides to send out an alarm signal to the appropriate alarm node. This kind of system may create latency in generating the alarm signal, particularly if the server is not local, and if any component of the wireless chain of communication is interrupted (e.g., the hub loses power), the alarm signal may never reach the alarm node and thus no alarm occurs. In one embodiment, multiple modes of communication may be used to reduce or eliminate these issues. For example, in addition to a first wireless communication protocol between the merchandise security devices **14** and gateway **24** and/or alarm nodes **30** and the gateway (e.g., WiFi, LoRa, etc.), a second wireless communication protocol may be used that is a direct node-to-node communication scheme between the merchandise security devices and the alarm nodes that does not have to also communicate with any hub or gateway. The communication protocols could be the same or different in some embodiments. In one example, the second wireless communication protocol could be performed using the same radio antennas that the other operational signals are communicated with the hub or gateway **24** (e.g., Wi-Fi, LoRa, etc.), which thereby adds no additional cost or size to either the merchandise security devices **14** and the alarm nodes **30** in order to accomplish the communication. However, a second radio is also an option. Additionally, the alarm signal could be broadcast on a different frequency than the other signals in order to address regional regulatory requirements

and/or if it is detected or known that certain frequency bands are getting congested. This communication could be two-way, but one-way communication would be sufficient in most circumstances. The merchandise security device **14** may send out a “help me” signal in response to a security event. The alarm node **30** would then only have to “listen” for that signal and if it receives the signal, the alarm node may generate an alarm by whatever means it is programmed for (e.g., light, sound, vibration, etc.).

(109) In some instances, a plurality of alarm nodes **30** may be used, and particular merchandise security device(s) **14** may be configured to activate specific alarm node(s). For example, in the instance where a retail store includes a plurality of display tables for a plurality of merchandise security devices **14**, there may be an alarm node **30** associated with each table which would only be triggered by a “help me” signal from any one of the merchandise security devices associated with the same table. In this situation, an identifier (e.g., an ID code) could be added to the “help me” signal that corresponds to a code stored in the alarm node **30**. Thus, the alarm node **30** may have to receive or identify its code in order to generate an alarm signal. This could be as simple as the code itself being the “help me” signal or some other instruction code could be added to or included with the identifier, for example, if more than one action (e.g., “alarm” or “stop alarming”) needed to be communicated to the alarm node. The merchandise security device **14** may be configured to generate this “help me” signal immediately upon a breach and only after sending the signal to the alarm node **30**, would the merchandise security device then communication via the wireless communication to a hub and gateway that a breach has occurred. Thus, the latency delay should be minimized in such a breach scenario.

(110) As discussed above, electronic keys **12**, **120** and computing devices **26** may be configured to communicate and/or control various security devices **14**. FIG. **43** illustrates embodiments of a merchandise display security system **200** include locks **202** used for locking various types of fixtures, such as cabinets and drawers. In the examples shown in FIGS. **43** and **47**, locks **202** may be used to secure sliding glass doors and drawers (see also FIGS. **48-49**). The system **10** may include various wireless functionality for communication between the locks **202**, computing devices **26**, hubs or gateways, electronic keys **12**, **120**, and/or remote devices. For instance, FIG. **44** illustrates that a retail store may include wireless communication circuitry in the form of a wireless router or other like hub **24** may facilitate Wi-Fi communication, although other forms of communication could be used such as cellular. The hub **24** may be used to facilitate communication between the computing devices **26** and one or more remote devices. In some cases, the electronic keys **120** may be configured to communicate with the one or more remote devices as well via the hub **24**. Communication between the computing devices **26** and one or more remote devices may be used to assigning authorization to the various computing devices and/or communicating various types of data such as the types of data disclosed above.

(111) Computing devices **26** may include wireless communications circuitry configured for BLE, Bluetooth, and/or NFC communication. The computing devices **26** may also or alternatively include a camera or a scanner for scanning images or information from the locks **202** as discussed in further detail below. Similarly, the locks **202** may include various wireless communications circuitry configured for BLE, Bluetooth, and/or NFC communication. The locks **202** may also or alternatively include a barcode or other identifier. In some cases, the computing devices **26** may be configured to be paired with one or more locks **202** (e.g., via Bluetooth communication) and/or include one or more additional communication protocols for operating the lock (e.g., NFC, camera, barcode, etc.).

(112) In one example embodiment, the computing devices **26** are configured to communicate with one or more locks **202** using a first communication protocol (e.g., Bluetooth). In order to unlock a specific lock, the computing device **26** may further be configured to communicate with each lock using a second communication protocol (e.g., NFC or image scanning). The second communication protocol may be used to identify a specific lock **202** that the computing device **26** is authorized to

unlock. For instance, an NFC tag may have an identifier that is unique to the lock **202** (similar to a serial number), and if the computing device **26** confirms that the identifiers match, then the computing device is authorized to unlock the lock. If the computing device **26** is authorized based on confirmation of identification of the lock **202**, the computing device may then communicate an unlock command to the lock using the first communication protocol.

(113) The locks **202** may take many different forms and configurations. The locks **202** may include various types of lock assemblies for different applications, such a plunger lock for sliding cabinet doors or a cam lock for drawers. FIG. **45** shows one embodiment of a lock **202**, where the lock includes a lock assembly, a drive assembly, an NFC tag, a transfer port with an IR transceiver, an inductive coil, a PCBA **214** with a Bluetooth module, and an internal power source (e.g., batteries). Moreover, FIG. **46** shows that the locks **202** may have different shapes depending on the application. For instance, some locks **202** may or may not include an internal power source, thereby affecting the size of the lock. In some applications, the internal power source may be external to the lock **202**, such as for a drawer where the lock may be positioned on the front of the drawer and the internal power source may be positioned inside the drawer and in electrical communication with the lock. In one embodiment further illustrated in FIG. **50**, the lock **202** may include an NFC tag **204** and a transfer port **206**, where the transfer port is similar to that described above for communication with an electronic key **12**, **120**. The NFC tag **204** may be positioned behind a cover **208** that masks or otherwise conceals the NFC tag. For instance, the cover may be plastic with a spun metal effect. In another example, the lock **202** may include a 2D barcode **210**. The lock **202** may include a removable cover **208** that is configured to conceal the NFC tag **204**, barcode **210**, or like identifier and to be removed for communication with a computing device **26**.

(114) As noted above, the lock **202** may be configured to communicate with an electronic key **120** for unlocking the lock. FIG. **51** shows an example of a key **120** communicating with the lock **202** via the transfer port **206**. The key **120** may be used in addition or alternatively to using a computing device **26** to unlock the lock. In the instance where the power source of the lock **202** is no longer capable of unlocking the lock (e.g., the batteries are depleted), the key **120** may be configured to transfer power to the lock for operating the lock, as disclosed above. In another embodiment, FIG. **52** shows that the internal power source may be a modular component **212** such that the power source may be replaced with another power source, such as in the form of a removable battery pack having a housing containing one or more batteries. In other cases, the removable battery pack may be removed and replaced with a cover if the internal power source is no longer needed or the lock is being used for a different application. Thus, embodiments of the present invention enable operation of the locks **202** even if the internal power source is incapable of unlocking the lock.

(115) In some embodiments, the modularity of the power source (e.g., battery pack) may be dependent or independent of the operation of the lock **202**. In this regard, theft of the power source may be problematic if it hinders the operation of the lock **202**. In one example, the locking mechanism used to unlock the lock **202** may be dependent on a mechanism for accessing the internal power source. Thus, a user would need to use a computing device **26** or electronic key **120** to access the internal power source. The lock **202** may be required to be in an unlocked state before the internal power source may be accessed thereby requiring an authorized user to be present before being able to access the internal power source. In other embodiments, a second lock mechanism that is independent of the locking mechanism of the lock **202** may be employed for accessing the internal power source. The second lock mechanism may be configured to be operated by a computing device **26**, electronic key **120**, and/or other type of key. For example, a mechanical lock mechanism may be operable using a magnetic key or tool configured to unlock the lock mechanism for releasing or accessing the internal power source. In some cases, different user access levels may be used such that only certain users are authorized to unlock the second lock mechanism for accessing the internal power source (e.g., a manager may be assigned access privileges for such access but a retail associate is not). Such access levels could be used when

assigning access privileges as disclosed above.

(116) In operation, FIG. 53 shows an example of a user using a computing device 26 to unlock a lock using NFC communication where the user places the computing device in close proximity to the NFC tag 204 which results in automatically unlocking the lock. FIG. 53 also shows that a user may use a camera or scanner of a computing device 26 to scan a barcode 210 for unlocking the lock. Consumers or store associates may use the camera of the computing device 26 to unlock the lock 202, whereas only a store associate may be authorized to use a scanner of a computing device 26. The computing device 26 may include a software application that facilitates communication with the locks in any of the above examples, such as by allowing a user to select an “unlock” command for unlocking the lock 202 if the user is authorized to do so. Authorization may be accomplished in various ways, such as via the embodiments described above (e.g., assignment of particular locks or zones). In other cases, the user may be authorized by virtue of being pre-authorized by downloading the software application and entering various information for identifying the user. The software application may also be password protected for ensuring the user is authorized to operate the lock 202. In addition, the software application may facilitate data collection and communication to one or more remote device.

(117) In some embodiments, the user may be required to manually unlatch the lock 202 after using a computing device 26 or electronic key 120 to unlock the lock. Following a successful unlock command from a computing device 26, FIG. 54 shows that the user may have a limited or pre-determined amount of time in which to unlatch the lock 202. For instance, the lock 202 may include a visible indicator (e.g., an LED) that illuminates or flashes different colors or frequencies depending on whether the lock 202 is capable of being unlatched or not. If the user chooses to unlatch the lock 202 after a successful unlocking command, the lock may be configured to be manually unlatched, such as by rotating or pulling a portion of the lock. For example, if the lock 202 is a cam style lock, the user may be able to rotate a knob for unlatching the lock, whereas if the lock is a plunger style lock, the user may be able to pull the knob for unlatching the lock. The lock 202 may be configured to automatically relock itself after a predetermined period time (e.g., 2-10 seconds). Moreover, the user may be required to manually relatch the lock 202. In some cases, the user may be required to rotate or push the knob of the lock 202 in an opposite direction to relatch that was used to unlatch the lock. If the user prematurely relatches the lock 202, the user may be required to first unlock the lock the lock to again relatch the lock when the fixture is in its fully closed position. It is understood that the lock 202 may include various actuators for unlatching the lock, such as knobs, handles, etc. that may be used to manually unlatch and relatch the lock. In other embodiments, a separate latching operation may be omitted, such as where the user is able to open the door without having to unlatch a latch mechanism.

(118) FIGS. 55-56 illustrate a lock 302 according to one embodiment of the present invention. In this embodiment, the lock 302 may be a cam lock configured for use with a fixture such as a sliding drawer. For example, unlocking of the lock 302 allows unlatching of a cam mechanism in engagement with the fixture to thereby allow access to the fixture. Similar to the embodiments discussed above, the lock 302 may be configured to communicate with various computing devices 26, hubs or gateways, electronic keys 12, 120, and/or remote devices. Moreover, the lock 302 may include a NFC tag 204, barcode 210, or like identifier for communication with a computing device 26, and/or a transfer port 206 for communication with an electronic key 12, 120. In this embodiment, a knob 304 may be configured to be turned by a user between latched and unlatched positions, although other actuations could be used (e.g., push/pull). The knob 304 may be coupled to a drive shaft 306 for rotating a cam mechanism (not shown) between latched and unlatched positions relative to a fixture.

(119) FIGS. 57-58 show embodiments of internal views of the lock 302 with the top of the lock housing removed for purposes of illustration. The lock 302 includes a worm gear 308 that is coupled to a motor 310 and a cam sleeve 312. Thus, energizing the motor 310 causes the worm

gear **308** to rotate which in turn causes the cam sleeve **312** to rotate. The cam sleeve **312** may be circular or ring-shaped with an internal cam surface **314** and an outer geared surface. The rotational axis of the cam sleeve **312** may be co-axial to the axis of the drive shaft **306**. In some cases, the cam sleeve **312** may surround the drive shaft **306**. The outer geared surface of the cam sleeve **312** is configured to mate with and engage the worm gear **308**. A pin **316** or other engagement member is configured to move between locked and unlocked states relative to the drive shaft **306** in response to rotation of the cam sleeve **312**. For example, an authorized user of a computing device **26** or electronic key **12, 120** may communicate with the lock **302** for moving the pin **316** to an unlocked state thereby disengaging the drive shaft **306**. The pin **316** may be biased towards a locked state with a spring or other biasing member and may be configured to move perpendicular to the axis of the drive shaft **306**. Thus, rotation of the worm gear **308** causes rotation of the cam sleeve **312**, which in turn causes the internal cam surface **314** to contact the pin **316**, which then causes the pin to retract out of engagement with the drive shaft **306** to an unlocked state.

(120) As shown in FIGS. **60-61**, the drive shaft **306** may include an opening **320** configured to receive the pin **316** therein in the locked state, although other forms of engagement may be used. In the unlocked state, the knob **304** is configured to rotate the drive shaft **306**, and in turn the cam mechanism that is in engagement with the fixture, to the unlatched position to thereby allow the fixture to be opened and accessed. The lock **302** may further include a switch **318** that is configured to be engaged and disengaged in response to rotation of the cam sleeve **312** for signaling the motor **310** to turn on or off. In this way, the motor **310** may turn on when an authorized computing device **26** or electronic key **12, 120** is presented and turn off when the cam sleeve **312** has rotated a predetermined rotational angle sufficient to disengage the pin **316** from the drive shaft **306**. In order to relatch the pin **316** with the drive shaft **306**, the knob **304** is configured to be rotated back to its initial position, and the pin may be biased in such a way that the pin automatically engages the drive shaft when the drive shaft is rotated back to its initial position. Similar to embodiments discussed above, the lock **302** may include a power source **322**, which may be housed within a modular component **212** in some cases (see, e.g., FIGS. **61-62**).

(121) With respect to installation, the lock **302** may include a modular adapter **324** in some embodiments. The modular adapter **324** may allow for installation of the lock **302** in different sized holes and in different orientations. Thus, the drive shaft **306** is configured to mate with different sized and configured modular adapters **324** used to mount the lock **302** to the fixture. The modular adapter **324** may include a threaded exterior surface and/or one or more flats to facilitate installation in the fixture. As shown in FIGS. **59-61**, for example, the modular adapter **324** may be configured to be positioned over the drive shaft **306** and secured in place with one or more fasteners **326** (e.g., screws, washers, and/or nuts). A cam mechanism for engaging and disengaging the fixture may be mounted at the end of the drive shaft **306** and adjacent to the modular adapter **324**.

(122) The foregoing has described one or more exemplary embodiments of various security systems. Embodiments of a security system have been shown and described herein for purposes of illustrating and enabling one of ordinary skill in the art to make, use and practice the invention. Those of ordinary skill in the art, however, will readily understand and appreciate that numerous variations and modifications of the invention may be made without departing from the spirit and scope thereof. Accordingly, all such variations and modifications are intended to be encompassed by the appended claims.

Claims

1. A security system for a fixture comprising: at least one lock configured to protect one or more items from theft from the fixture, wherein the lock comprises a drive shaft configured to be moved between a latched position and an unlatched position, the drive shaft comprising an engagement

member configured to be in engagement with the fixture in the latched position and to be disengaged from the fixture in the unlatched position, wherein the lock is configured to be moved between a locked state and an unlocked state, the engagement member being in engagement with the fixture, and the drive shaft being unable to be moved to the unlatched position, when in the locked state, wherein the drive shaft is configured to be rotated from the latched position to the unlatched position to thereby move the engagement member out of engagement with the fixture when in the unlocked state for accessing the fixture, wherein the lock comprises a cam sleeve having an internal cam surface configured to transition the lock between the locked state and the unlocked state in response to receiving a wireless authorization signal.

2. The security system of claim 1, further comprising a mobile computing device configured to wirelessly communicate the wireless authorization signal to the lock.
3. The security system of claim 2, further comprising an electronic key configured to wirelessly communicate the wireless authorization signal to the lock.
4. The security system of claim 1, wherein the drive shaft is configured to be rotated between the latched position and the unlatched position.
5. The security system of claim 1, wherein the cam sleeve is configured to be rotated to transition between the locked state and the unlocked state.
6. The security system of claim 1, further comprising an engagement member configured to be moved into and out of engagement with the drive shaft in response to movement of the cam sleeve, the drive shaft being in the locked state when the engagement member is engaged with the drive shaft and in the unlocked state when the engagement member is disengaged from the drive shaft.
7. The security system of claim 6, wherein the drive shaft is configured to rotate about an axis, and wherein the engagement member is configured to move perpendicular to the axis.
8. The security system of claim 6, wherein the engagement member is a pin, and wherein the drive shaft defines an opening configured to receive the pin therein.
9. The security system of claim 6, wherein the engagement member is biased towards the locked state.
10. The security system of claim 1, wherein the cam sleeve is ring shaped.
11. The security system of claim 1, wherein the cam sleeve is co-axial to the drive shaft.
12. The security system of claim 1, wherein the lock further comprises a worm gear in engagement with the cam sleeve and configured to move the cam sleeve.
13. The security system of claim 12, further comprising a motor operably engaged with the worm gear and configured to rotate the worm gear when activated.
14. The security system of claim 1, wherein the lock further comprises a knob coupled to the drive shaft and configured to be manually actuated for rotating the drive shaft between the latched position and the unlatched position.
15. The security system of claim 1, wherein the lock further comprises a cam mechanism coupled to the drive shaft and configured to engage and disengage the fixture in response to movement of the drive shaft between the latched position and the unlatched position.
16. The security system of claim 1, wherein the lock does not have an internal power source.
17. The security system of claim 1, wherein the lock includes an internal power source.
18. The security system of claim 17, wherein the lock comprises a housing containing the internal power source, and wherein the housing is modular and configured to be attached and detached from the lock.
19. The security system of claim 1, wherein the drive shaft and the cam sleeve are configured to rotate about the same axis.
20. The security system of claim 1, wherein the lock is configured to be mounted to an outer surface of the fixture, and wherein the drive shaft is configured to extend through an opening defined in the fixture to an inner surface of the fixture, opposite the outer surface.
21. The security system of claim 1, wherein the drive shaft is configured to be manually rotated

between the latched position and the unlatched position.

22. The security system of claim 1, wherein the drive shaft is configured to be rotated from the unlatched position to the latched position for engaging the engagement member with the fixture.

23. The security system of claim 1, wherein the engagement member is configured to be in engagement with the fixture when in the unlocked state.

24. A method for securing items from theft from a fixture, the method comprising: providing at least one lock configured to protect one or more items from theft from the fixture, wherein the lock comprises a drive shaft and a cam sleeve having an internal cam surface configured to transition the lock between a locked state and an unlocked state in response to movement of the cam sleeve; causing a wireless authorization signal to be transmitted to the lock for moving the cam sleeve and thereby transition the lock from the locked state to the unlocked state, the drive shaft being in engagement with the fixture and unable to be moved to the unlatched position when in the locked state; and rotating the draft shaft from a latched position to an unlatched position while the lock is in the unlocked state for accessing the fixture, the drive shaft comprising an engagement member configured to be in engagement with the fixture in the latched position and to be disengaged from the fixture in the unlatched position.
